

Driver Drowsiness Detection Using Eye Closure and Yawning Analysis with Deep Learning

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Abstract

Driver drowsiness is a major contributor to road accidents worldwide, resulting in reduced reaction time, impaired judgement, and decreased situational awareness. This paper presents a vision-based driver drowsiness detection system using deep learning techniques. Facial indicators such as eye closure and yawning are analysed to detect fatigue levels. Two deep learning approaches, namely a Custom Convolutional Neural Network (CNN) and MobileNetV2 Transfer Learning model, were implemented and evaluated. The system classifies facial images into four categories: Open, Closed, Yawn, and No Yawn. A decision fusion mechanism is further employed to classify fatigue into three stages: Alert, Mild Fatigue, and Severe Fatigue. Experimental results demonstrate that MobileNetV2 achieves superior accuracy, faster convergence, and better generalisation performance compared to the Custom CNN model.

Keywords: Driver Drowsiness Detection, Deep Learning, Computer Vision, CNN, MobileNetV2, Transfer Learning, Fatigue Monitoring.

I. INTRODUCTION

Driver fatigue is one of the leading causes of traffic accidents and safety-related incidents. Long driving hours, insufficient sleep, and medical conditions contribute to drowsiness, resulting in slower reaction times and impaired decision-making.

Traditional fatigue detection systems often rely on wearable sensors or vehicle behaviour analysis. These methods can be intrusive, costly, and less reliable under varying driving conditions. Recent advancements in computer vision and deep learning have enabled non-intrusive fatigue detection systems using facial analysis.

This project focuses on identifying fatigue through eye closure and yawning detection using image classification techniques.

II. PROBLEM STATEMENT

The primary challenge is to develop a robust vision-based driver monitoring system capable of identifying early signs of driver fatigue through facial indicators.

The system should:

- Detect eye states (Open or Closed)
- Detect mouth states (Yawn or No Yawn)
- Classify fatigue levels
- Monitor fatigue progression over time
- Operate efficiently for future real-time deployment

III. OBJECTIVES

The objectives of this study are:

1. To preprocess and organise facial image datasets.
2. To develop a Custom CNN model.
3. To implement MobileNetV2 Transfer Learning.
4. To compare both models using performance metrics.
5. To design a fatigue-stage classification system.
6. To analyse driver fatigue progression over time.

IV. LITERATURE REVIEW

Several studies have demonstrated that facial indicators such as eye blinking frequency, prolonged eye closure, and yawning provide reliable information regarding driver alertness.

Deep learning methods offer significant advantages over traditional machine learning techniques because feature extraction is performed automatically through neural network architectures.

Transfer learning models such as MobileNetV2 have gained popularity due to their ability to achieve high classification accuracy while maintaining computational efficiency.

V. DATASET DESCRIPTION

The dataset consists of facial images collected for fatigue detection.

Classes

Class	Description
Open	Eyes Open
Closed	Eyes Closed
no yawn	No Yawning
yawn	Yawning

Dataset Structure

The dataset was organised into:

- Training Set
- Validation Set
- Test Set

To ensure dataset integrity, SHA-256 hash verification was performed to confirm that duplicate images were not present across multiple partitions.

VI. DATA PREPROCESSING

Data preprocessing plays a crucial role in improving model performance.

Image Resizing

All images were resized to:

Plain Text
224 × 224 pixels
Show more lines

Normalisation

For the Custom CNN:

Python
rescale = 1./255

Show more lines

For MobileNetV2:

Python

```
preprocess_input()
```

Show more lines

Data Augmentation

The following augmentation techniques were applied:

Python

```
rotation_range=10
```

```
zoom_range=0.1
```

```
horizontal_flip=True
```

```
brightness_range=(0.8,1.2)
```

Show more lines

These methods increase data diversity and reduce overfitting.

VII. PROPOSED METHODOLOGY

The proposed framework consists of the following stages:

Plain Text

Dataset Collection

↓

Data Preprocessing

↓

Data Augmentation

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Custom CNN Training

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MobileNetV2 Training

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Performance Evaluation

↓

Fatigue Classification Logic

↓

Fatigue Progression Analysis

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Show more lines

VIII. CUSTOM CNN ARCHITECTURE

A Custom CNN model was developed using multiple convolutional and pooling layers.

Architecture

Plain Text
Input Layer
↓
Conv2D (32)
↓
MaxPooling
↓
Conv2D (64)
↓
MaxPooling
↓
Conv2D (128)
↓
MaxPooling
↓
Flatten
↓
Dense (128)
↓
Dropout (0.4)
↓
Softmax Output
[Show more lines](#)

Training Configuration

- Optimizer: Adam
- Loss Function: Categorical Cross entropy
- Epochs: 15
- Batch Size: 32

IX. MOBILENETV2 TRANSFER LEARNING MODEL

MobileNetV2 was selected due to its lightweight architecture and strong generalisation capability.

Architecture

Plain Text
MobileNetV2 Backbone
↓
GlobalAveragePooling
↓
Dropout
↓
Dense Output Layer
Show more lines

Transfer Learning Strategy

The pretrained ImageNet backbone was frozen:

Python
backbone.trainable = False
Show more lines

Only the classifier layers were trained on the fatigue dataset.

X. MODEL TRAINING

Both models were trained with identical data partitions.

Hyperparameters

Parameter	Value
Image Size	224×224
Batch Size	32
Epochs	15
Optimizer	Adam
Loss Function	Categorical Cross entropy
Early Stopping	Enabled

Early stopping was employed to prevent overfitting.

XI. PERFORMANCE EVALUATION

Model performance was evaluated using the following metrics:

Accuracy

Measures overall classification performance.

Precision

Measures correctness of positive predictions.

Recall

Measure's ability to identify actual positives.

F1-Score

Provides a balanced measure between Precision and Recall.

Confusion Matrix

Used to analyse class-wise prediction performance.

Insert Accuracy Graph Here

Insert Loss Graph Here

Insert Confusion Matrix Here

XII. ERROR ANALYSIS

Misclassified samples were examined to identify model limitations.

Common Error Cases

Eye-State Classification

- Partially closed eyes
- Reflections from spectacles
- Motion blur effects

Mouth-State Classification

- Small mouth openings
- Facial occlusions
- Non-frontal facial orientations

Observations

Most errors occurred under poor illumination conditions and ambiguous facial expressions.

XIII. FATIGUE DECISION FUSION

The original four-class predictions were converted into three fatigue stages.

Prediction	Fatigue Stage
Open	Alert
No Yawn	Alert
Yawn	Mild Fatigue
Closed	Severe Fatigue

Implementation:

```
Python
Stage for_class = {
"Open":0,
"No_yawn":0,
"yawn":1,
"Closed":2
}
Show more lines
```

This mapping improves interpretability for real-world monitoring applications.

XIV. FATIGUE PROGRESSION ANALYSIS

To understand changes in driver alertness over time, predictions were grouped into sequential intervals and converted into fatigue stages.

The progression curve illustrates transitions between:

- Alert
- Mild Fatigue
- Severe Fatigue

Insert Fatigue Progression Curve Here

XV. RESULTS AND DISCUSSION

Model Comparison

Metric	Custom CNN	MobileNetV2
Training Accuracy	86%	89%
Validation Accuracy	84%	85%
Test Accuracy	82%	85%
Training Time	9 min	8 min

Discussion

The MobileNetV2 model demonstrated superior performance due to the benefits of transfer learning. It achieved faster convergence, better validation accuracy, and stronger generalisation on unseen data.

XVI. LIMITATIONS

The current system has the following limitations:

- Sensitivity to poor lighting conditions
 - Limited side-face robustness
 - Dependence on image quality
 - Lack of temporal sequence modelling
-

XVII. FUTURE WORK

Future enhancements may include:

- Real-time webcam integration

- Audio warning systems
 - Eye Aspect Ratio (EAR) based detection
 - Face landmark analysis
 - LSTM-based temporal modelling
 - Mobile and edge-device deployment
 - Advanced Driver Assistance System integration
-

XVIII. CONCLUSION

This paper presented a deep learning-based driver drowsiness detection system using eye closure and yawning analysis. Both a Custom CNN and MobileNetV2 model were developed and evaluated. A fatigue classification framework was implemented to categorise drivers into Alert, Mild Fatigue, and Severe Fatigue stages. Experimental observations indicate that MobileNetV2 provides better accuracy, faster training, and improved generalisation performance. The developed system demonstrates the effectiveness of non-intrusive vision-based fatigue detection and can be extended for real-time intelligent transportation applications.

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